Production Order Create/Edit — Basic Design Document (基本設計書)

> **Revision of 2026-10-24 (WI-017).** This file supersedes `001_BD_製造指示登録・編集.md`, which stays unchanged as the earlier approved revision. Change: the production-order notes limit is 1000 characters (REQ-090, supersedes the 500 of REQ-016); see `work-items/WI-017/`.

Document control (改版履歴)

Field	Value
Document ID	001_BD
Category	UI
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-002
Based on brief.md revision	1
Created by	Claude (for ThanhTN)
Created date	2026-09-18
Last updated by	Claude (for ThongTM)
Last updated date	2026-10-24

Version	Date	Author	Revision content
1	2026-09-18	Claude (for ThanhTN)	Initial creation (WI-002 restart against the rewritten BD template; supersedes the scrapped first-pass 001_BD)
2	2026-09-18	Claude (for ThanhTN)	DEC-009–DEC-012 resolved: due-date check condition, stale-save rejection (V-08), plant timezone, `PO-YYYY-NNNNN` order number
3	2026-09-18	Claude (for ThanhTN)	DEC-017–DEC-020: plant timezone `Asia/Tokyo`; Cancel confirmation when edited (REQ-019, items 17–19, E-07/E-09); 30 seeded products; CSRF approach decided
4	2026-09-18	Claude (for ThanhTN)	Aligned with 001_DD: quantity upper bound (V-02, DEC-024)

Version	Date	Author	Revision content
5	2026-09-22	Claude (for ThanhTN)	Screen B exists (WI-003): the screen's entry point and every exit that used to lead to the home page now lead to the production-order list
6	2026-09-22	Claude (for ThanhTN)	Completion tracking for Screen C (WI-004 REQ-033, DEC-002/DEC-003): saving `InProgress → Completed` records the completion time (FN-006, Actions and business rules, Data design overview). No visible change to SCR-001
7	2026-09-22	Claude (for ThanhTN)	WI-004 (DEC-016, DEC-021): the shared header now carries the application navbar specified in 003_BD "Shared application header"; items 1–2 point to it. The breadcrumb stays. WI-004 DEC-022: the discard-changes confirmation (FN-009, dialog 17) now also guards leaving an edited form through any in-app link — navbar, breadcrumb, app name — not only Cancel (E-07a); Discard then goes to that link's destination. No field, rule, validation or API changes
8	2026-09-23	Claude (for ThanhTN)	Diagrams redrawn for the revised BD template (RFC 0009): screen transition and status state machine in Mermaid; PC and SP layouts as SVG wireframes under `wireframes/`. The screen transition now matches the screen list: Cancel returns to SCR-002, which the text diagram had not shown since version 5. No field, rule, validation, event or API changes
9	2026-09-23	Claude (for ThanhTN)	WI-005: the UI is Japanese. Quoted UI text and messages give the implemented Japanese text with an English gloss (WI-005 DEC-003); status labels (M-01, M-02), timestamp format (M-04, `YYYY/MM/DD HH:mm`), page titles and wireframes updated; demo products are automobile parts with Japanese names; two screen-transition edges drawn longer so their labels don't overlap in the PDF. No field, rule, validation, event or API changes
10	2026-10-24	Claude (for ThongTM)	WI-017 (REQ-090–REQ-092): notes limit 500 → 1000 characters. Requirement trace, items 11–12, V-05 and the MSG-E006 text updated; wireframes replaced by dated copies showing `0/1000`. No other field, rule, event or API change. Dated file name per WI-017 DEC-002

System overview

One screen, SCR-001, lets a signed-in Admin or Operator create a new production order or edit an existing one, including moving it through its status workflow. It is the harness's first end-to-end business screen ("Screen A").

Requirement ID	Description	Covered by section
REQ-010	Create a production order	Business flow; §3 screen items; §6 events; Success and exception flows
REQ-011	Edit an existing production order	Business flow; 0-3 URL parameters; §5 V-08; §6 events; Success and exception flows
REQ-012	Only authenticated Admin/Operator	0-1 basic information; Actions and business rules; Exception flows
REQ-013	Quantity positive whole number	§5 validation V-02
REQ-014	Due date today or later	§5 validation V-03, V-04
REQ-015	Product must exist	§5 validation V-01; Data design overview
REQ-016, REQ-090	Notes ≤ 1000 characters (REQ-090 supersedes REQ-016's 500)	§5 validation V-05

Requirement ID	Description	Covered by section
REQ-017	Fixed status state machine	Screen transition (status diagram); §3 item 6; §4 M-02; §5 V-06
REQ-018	Product/quantity locked after Draft	§3 display conditions; §5 V-07; Exception flows
REQ-019	Confirm before discarding changes on Cancel	§3 items 17–19; §6 E-07, E-09

Overall configuration and architecture

Uses the confirmed stack (`ai/project.md`): a React (Vite + TypeScript + Tailwind CSS v4) page calling the .NET 10 backend's JSON API over the same origin, layered Domain/Application/Infrastructure/Api (0001_ADR), persisted in PostgreSQL 17. Access is gated by WI-001's cookie session and role checks (0002_ADR): `ProtectedRoute` on the frontend, an authorization policy on the backend. No new architecture boundary or external system is introduced.

Function list

Function ID	Function name	Description	Related requirement ID
FN-001	Create production order	Save a new order with a system-generated order number and status `Draft`	REQ-010
FN-002	Edit production order	Save changes to an existing order's editable fields	REQ-011, REQ-018
FN-003	Load production order	Fetch an existing order for display in edit mode	REQ-011
FN-004	List products	Provide the product choices for the product dropdown	REQ-015
FN-005	Validate order input	Enforce product, quantity, due date and notes rules (client for feedback, server authoritative)	REQ-013, REQ-014, REQ-015, REQ-016
FN-006	Enforce status transitions	Allow only the defined state-machine edges; on `InProgress → Completed` , record the completion time (WI-004 FN-021)	REQ-017; WI-004 REQ-033
FN-007	Enforce field lock by status	Reject product/quantity changes once the order is not `Draft`	REQ-018
FN-008	Enforce authentication and role	Reject unauthenticated (401) and non-Admin/Operator (403) requests	REQ-012
FN-009	Confirm discard on leaving	Ask before discarding unsaved changes, whether the user leaves by Cancel or by an in-app link (WI-004 DEC-022)	REQ-019

Actors and business flow

Actors: Admin, Operator — both already signed in through WI-001's login screen (DEC-001).

Create (UC-001)

1. User opens the create route → system loads the product list (FN-004) and shows an empty form; status shows `Draft`, order number shows 「保存時に採番」 (Assigned on save).
2. User selects a product and enters quantity, due date and optional notes → system shows field errors as fields lose focus (FN-005).
3. User presses **保存** (Save) → system validates all fields client-side; if valid, sends the request; the server re-validates (FN-005, FN-008) and saves (FN-001).
4. On success → system navigates to the edit route of the new order and shows 「製造指示 {order number} を登録しました。」 (Production order {order number} created.)

Edit (UC-002, UC-003)

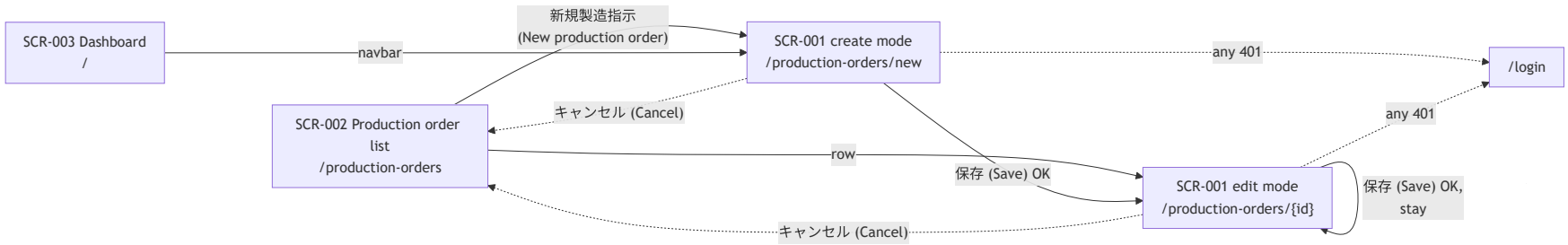
1. User opens the edit route for an order → system loads the order (FN-003) and products (FN-004) and shows the form pre-filled; product/quantity are read-only unless status is `Draft`; the status dropdown offers only the current status and its valid next states (DEC-008).
2. User changes editable fields and/or status → client-side validation as above.
3. User presses **保存** (Save) → server re-validates (FN-005–FN-008) and saves (FN-002).
4. On success → form shows the saved values and 「製造指示 {order number} を保存しました。」 (Production order {order number} saved.)

Cancel (REQ-019): **キャンセル** (Cancel) returns to the production-order list, SCR-002 (`/production-orders`), without saving. If the form has unsaved changes, the system first asks 「変更を破棄しますか？」 (Discard your changes?) (FN-009); **編集を続ける** (Keep editing) returns to the form with all values kept.

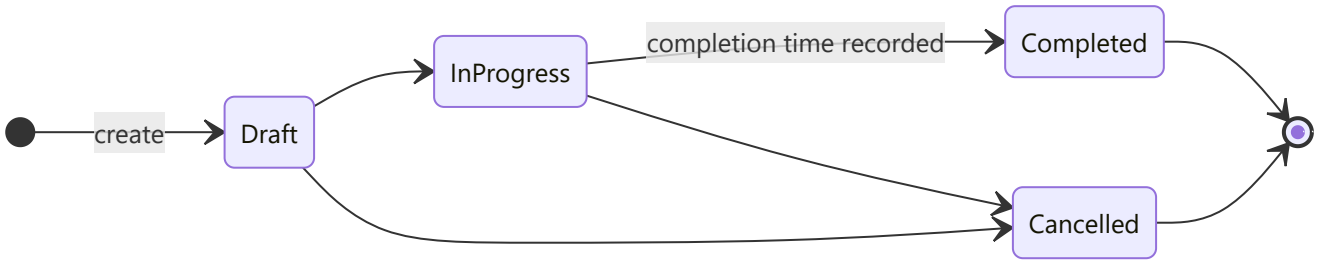
Screen list and screen transition

Screen ID	Screen name	Entry point	Exit / next screen
SCR-001	Production Order Create/Edit	Create mode: `/production-orders/new`, from the 「新規製造指示」 (New production order) action on SCR-002 or on the home page. Edit mode: `/production-orders/{id}`, from a row on SCR-002 (or the direct route).	Save (create) → SCR-001 edit mode for the new order. Save (edit) → stays on SCR-001. Cancel → SCR-002 `/production-orders`. Session expired / 401 → `/login`.

Screen transition:



Order status state machine (REQ-017, DEC-003):



`Completed` and `Cancelled` are terminal.

Screen design detail

SCR-001 Production Order Create/Edit

0-1. Basic information (基本情報)

No	Item	Content	Reference
1	Route / path	`/production-orders/new` (create), `/production-orders/{id}` (edit)	0-3
2	API base path	`/api/production-orders` , `/api/products`	Exact contract in 001_DD

No	Item	Content	Reference
3	Character encoding	UTF-8	
4	Error page / fallback	Load failure or unknown `{id}`: in-page error panel with a link back to `/`; no separate error route	Exception flows
5	Responsive	Yes — PC (≥ 640px) and SP (< 640px, Tailwind `sm` breakpoint)	§1
6	Authentication required	Yes	REQ-012
7	Authorization / role restriction	`Admin`, `Operator` (DEC-001)	REQ-012
8	Applicable channel(s)	Single web app — not applicable	

0-2. Page metadata (head)

The app's common head (`src/frontend/index.html`: charset, viewport, favicon, bundled module script) is not restated.

0-2-1. Title

No	Title
1	Create mode: `新規製造指示 – ProductionManagementAI` (New production order)
2	Edit mode: `製造指示 {order number} – ProductionManagementAI` (Production order {order number})

0-2-2. Base

Not applicable — no `<base>` override.

0-2-3. Link

Not applicable — no screen-specific `<link>` tags; styles are bundled.

0-2-4. Meta

Not applicable — no screen-specific meta tags (internal, authenticated screen; no SEO metadata needed).

0-2-5. Style

Not applicable — no inline `<style>` block; all styling via Tailwind classes.

0-2-6. Script

Not applicable — no extra `<script>` tags; app JS is bundled.

0-3. URL parameters

No	Parameter name	Content	Required / Optional	Reference
1	`id` (path segment)	Production order identifier; present only in edit mode	required in edit mode; absent in create mode	Unknown/invalid `id` → not-found exception flow

1. Layout and mockup

Layout-level sketch only; the rendered per-state mockup belongs in 001_DD.

PC / desktop


1 ProductionManagementAI
ダッシュボード
製造指示一覧
+ 新規製造指示
2 Admin · ログアウト
3 ホーム › 製造指示一覧 › 新規
4 新規製造指示
5 製造指示 PO-2026-00001 を登録しました。
6 指示番号
保存時に採番
7 ステータス
下書き
8 製品 *
製品を選択
9 数量 *
10 納期 *
yyyy/mm/dd
11 備考
任意
12 0/1000
13 登録日時 2026/09/18 10:12
14 最終更新 2026/09/20 09:30
13–14: edit mode only
15 キャンセル
16 保存
Item 7 is a select in edit mode; 8 and 9 are read-only unless the status is Draft. Dialog 17–19 (discard changes) is not drawn.

Item No.	Region / element	Notes (behavior, condition)
1	App header with navbar	Shared application header, specified once in 003_BD "Shared application header" (H-1–H-5, E-29–E-31, WI-004 DEC-016): app name, navbar (「ダッシュボード」 (Dashboard), 「製造指示一覧」 (Production orders), 「新規製造指示」 (New production order); current entry marked), user and 「ログアウト」 (Sign out), a 「メニュー」 (Menu) button on SP. Current entry here: 「新規製造指示」 on create mode; 「製造指示一覧」 on edit mode
2	Signed-in user + 「ログアウト」 (Sign out)	Shared header (003_BD H-3; WI-001 behavior unchanged)
3	Breadcrumb	Last segment 「新規」 (New) (create) or the order number (edit)
4	Page heading	Text by mode, same as 0-2-1 minus the app suffix
5	Message banner	Hidden unless there is a success or form-level error message; announced to assistive tech
6–14	Form fields	See §3; form card max width ~ 720px, centered
15	「キャンセル」 (Cancel) button	Secondary style
16	「保存」 (Save) button	Primary style; disabled while saving; reads 「保存中…」 (Saving…) meanwhile
17–19	Discard-changes dialog	Modal, centered; shown only by E-07 when the form has changes (not drawn above)

SP / mobile



Same item numbers as PC. Differences: single column; quantity and due date stacked; buttons full width, 「保存」 above 「キャンセル」; breadcrumb collapses to a

back link; the navbar collapses behind the header's Menu button (003_BD H-4, H-5).

2. Content block definition (CMS)

None — no externally managed content blocks.

3. Screen item definition

No	Item (label)	Variable name	Control type	I/O	Data type	Width / length	Initial value	Placeholder	Display condition	Data source	Reference
4	Page heading	—	label	O	text	—	By mode (0-2-1)	—	Always	Mode + `orderNumber`	
5	Message banner	—	label	O	text	—	Hidden	—	After save success or form-level error	Save/load result	§6 E-05, E-06
6	「指示番号」 (Order number)	`orderNumber`	label	O	text	—	Create: 「保存時に採番」 (Assigned on save); edit: saved value	—	Always	ProductionOrder.order number	`PO-YYYY-NNNNN` (DEC-012)
7	「ステータス」 (Status)	`status`	select (edit) / label (create)	I/O	enum	—	Create: `Draft` (label); edit: saved value	—	Always; select disabled when status is `Completed` or `Cancelled`	ProductionOrder.status	§4 M-01, M-02; DEC-008
8	「製品」 (Product)	`productId`	select	I/O	identifier	—	Create: none selected; edit: saved value	「製品を選択」 (Select a product)	Always; read-only unless status is `Draft` (REQ-018)	Product list (FN-004)	§4 M-03
9	「数量」 (Quantity)	`quantity`	textbox (numeric)	I/O	integer	up to 9 digits	Create: empty; edit: saved value	—	Always; read-only unless status is `Draft` (REQ-018)	ProductionOrder.quantity	

No	Item (label)	Variable name	Control type	I/O	Data type	Width / length	Initial value	Placeholder	Display condition	Data source	Reference
10	「納期」 (Due date)	`dueDate`	date picker	I/O	date	—	Create: empty; edit: saved value	—	Always; editable in every status (DEC-004)	ProductionOrder.due date	
11	「備考」 (Notes)	`notes`	textarea	I/O	text	≤ 1000 chars	Create: empty; edit: saved value	「任意」 (Optional)	Always; editable in every status (DEC-004)	ProductionOrder.notes	
12	Notes counter	—	label	O	text	—	"0/1000"	—	Always	Live length of item 11	
13	「登録日時」 (Created)	`createdAt`	label	O	datetime	—	Saved value	—	Edit mode only	ProductionOrder.created timestamp	§4 M-04
14	「最終更新」 (Last updated)	`updatedAt`	label	O	datetime	—	Saved value	—	Edit mode only	ProductionOrder.updated timestamp	§4 M-04
15	「キャンセル」 (Cancel)	—	button	I	—	—	—	—	Always	—	§6 E-07
16	「保存」 (Save)	—	button	I	—	—	—	—	Always; disabled while a save is in progress	—	§6 E-04
17	Discard-changes dialog	—	modal dialog	O	text	—	Hidden	—	After Cancel when the form has unsaved changes	—	Title 「変更を破棄しますか？」 (Discard your changes?), body 「この製造指示の変更は保存されていません。」 (Your changes to this production order haven't been saved.); REQ-019
18	「破棄」 (Discard)	—	button	I	—	—	—	—	Inside dialog 17	—	§6 E-09

No	Item (label)	Variable name	Control type	I/O	Data type	Width / length	Initial value	Placeholder	Display condition	Data source	Reference
19	「編集を続ける」 (Keep editing)	—	button	I	—	—	—	—	Inside dialog 17; receives initial focus	—	§6 E-09

Required fields (8, 9, 10) are marked with `*` and `aria-required`, with the legend 「* は必須項目です。」 (* marks a required field). Once the order leaves `Draft`, items 8 and 9 carry the hint 「下書き以外の製造指示では変更できません。」 (Can't be changed once the order leaves Draft).

4. Item value mapping

No	Item	Source value	Displayed value	Reference
M-01	(7) Status	`Draft` / `InProgress` / `Completed` / `Cancelled`	「下書き」 / 「進行中」 / 「完了」 / 「取消」 (Draft / In progress / Completed / Cancelled)	WI-005 DEC-004
M-02	(7) Status options	Current status	`Draft` → {下書き, 進行中, 取消}; `InProgress` → {進行中, 完了, 取消}; `Completed` → {完了} (disabled); `Cancelled` → {取消} (disabled). Labels as M-01	REQ-017, DEC-008
M-03	(8) Product	Product ID	"{SKU} — {name}", e.g. `P-1004 – ドライブシャフト`	DEC-005; WI-005 DEC-002
M-04	(13)(14) Timestamps	UTC timestamp	`YYYY/MM/DD HH:mm`, 24-hour, in the browser's timezone, after the label: 「登録日時 2026/09/18 10:02」	WI-005 DEC-006

5. Validation rules

Client-side checks give early feedback; the server repeats every check and is authoritative. Error message texts are drafts; final IDs/wording are fixed in 001_DD.

No	Item	Check content	Validation rule	Check condition	Error message (ID)	Reference
V-01	(8) Product	Required; must be an existing product	Not empty; exists in Product	Create; edit while `Draft`	MSG-E001 「製品を選択してください。」 (Select a product.) / MSG-E002 「選択した製品は存在しません。」 (The selected product no longer exists.)	REQ-015
V-02	(9) Quantity	Required positive whole number	Integer 1–999,999,999	Create; edit while `Draft`	MSG-E003 「1以上の整数を入力してください。」 (Enter a whole number of 1 or more.) / MSG-E010 「数量は999,999,999以下で入力してください。」 (Quantity can't exceed 999,999,999.)	REQ-013, DEC-024

No	Item	Check content	Validation rule	Check condition	Error message (ID)	Reference
V-03	(10) Due date	Required valid date	Not empty; valid date	Always	MSG-E004 「納期を入力してください。」 (Enter a due date.)	REQ-014
V-04	(10) Due date	Today or later	≥ today, where "today" is the date in the configured plant timezone, `Asia/Tokyo` (DEC-011, DEC-017)	On create; on edit only when the due date differs from the saved value (DEC-009)	MSG-E005 「納期に過去の日付は指定できません。」 (Due date can't be in the past.)	REQ-014, DEC-009, DEC-011
V-05	(11) Notes	Length	≤ 1000 characters	Always	MSG-E006 「備考は1000文字以内で入力してください。」 (Notes can't exceed 1000 characters.)	REQ-016, REQ-090
V-06	(7) Status	Allowed transition	New status is the current status or a valid next state (M-02)	Edit	MSG-E007 「このステータス変更は許可されていません。」 (This status change isn't allowed.)	REQ-017
V-07	(8)(9) Product, Quantity	Locked fields unchanged	If saved status ≠ `Draft`, product and quantity must equal saved values; otherwise the whole request is rejected	Edit	MSG-E008 「下書き以外の製造指示では、製品と数量を変更できません。」 (Product and quantity can't be changed after the order leaves Draft.)	REQ-018, DEC-007
V-08	Whole order	Not changed by someone else since loaded	Saved version equals the version the form was loaded with	Edit (server only)	MSG-E009 「この製造指示は他のユーザーによって変更されました。再読み込みして最新の内容を確認してください。」 (This order was changed by someone else. Reload to see the latest version.)	REQ-011, DEC-010

6. Item events

No	Item	Event	Event content	Reference
E-01	Screen	load	Create: load products, show empty form. Edit: load order and products; on unknown `id` show not-found panel; on 401 go to `/login`; show a loading indicator until both loads finish	FN-003, FN-004
E-02	(8)(9)(10)(11) fields	blur	Run that field's client-side checks (§5) and show/clear its inline error	FN-005
E-03	(11) Notes	input	Update counter (12)	
E-04	(16) Save	click	Run all client checks; if any fail, show inline errors and move focus to the first invalid field; otherwise send the request, disable Save, and handle the result (E-05/E-06)	FN-001, FN-002
E-05	(16) Save — success	response	Create: navigate to `/production-orders/{new id}` and show success banner. Edit: refresh the form with saved values (including status options and lock state) and show success banner	REQ-010, REQ-011
E-06	(16) Save — failure	response	Field errors → inline under each field; form-level errors (transition, lock, not found, server error) → error banner (5); 401 → `/login`; entered values are kept	Exception flows

No	Item	Event	Event content	Reference
E-07	(15) Cancel	click	If any field differs from the loaded (edit) or initial (create) values, open dialog 17. Otherwise navigate to `/production-orders` (SCR-002) without saving	REQ-019, DEC-018
E-08	(7) Status	change	Only selectable options per M-02; no immediate save — takes effect on Save	DEC-008
E-07a	Navbar, breadcrumb or app-name link	click / Enter	If the form has unsaved changes (as E-07), cancel the navigation and open dialog 17, remembering the link's destination. Otherwise navigate normally. Browser Back, reload and closing the tab are not guarded (unchanged; WI-004 DEC-022)	REQ-019; WI-004 DEC-022
E-09	(18) Discard / (19) Keep editing	click / Escape	Discard: close the dialog and navigate — to the remembered destination when opened by E-07a, otherwise to `/production-orders` — without saving. Keep editing or Escape: close the dialog, return focus to Cancel, keep all values	REQ-019

7. External identity linkage

None — no external identity linkage.

Actions and business rules

Action	Trigger	Business rule	Related requirement ID
Open screen / call API	Any request to SCR-001 or its API	Requires a valid session (else 401 → `/login`) and role `Admin` or `Operator` (else 403) — enforced server-side, not only in the UI	REQ-012
Create order	Save in create mode	All V-01–V-05 pass; order gets a system-generated unique order number `PO-YYYY-NNNNN` (year of creation in the `Asia/Tokyo` plant timezone, sequence restarts yearly — DEC-012) and status `Draft`; created/updated timestamps set	REQ-010, REQ-013–REQ-016
Edit order	Save in edit mode	Applicable V-01–V-08 pass; only editable fields for the saved status are changed; updated timestamp set	REQ-011, REQ-013–REQ-018
Change status	Save with a different status	Only `Draft→InProgress`, `Draft→Cancelled`, `InProgress→Completed`, `InProgress→Cancelled`	REQ-017
Record completion	Save of `InProgress→Completed`	The order's completion time is set to the time of this save, in the same save as the status change; server-set, never entered or shown on SCR-001, never changed afterwards (`Completed` is terminal). A rejected save records nothing. Consumed by the dashboard, SCR-003 (003_BD)	WI-004 REQ-033
Change product/quantity	Save in edit mode	Allowed only while saved status is `Draft`; otherwise the whole request is rejected	REQ-018, DEC-007
Cancel / leave	Cancel button, or any in-app link	Unsaved changes are discarded only after the user confirms	REQ-019, DEC-018; WI-004 DEC-022

Success and exception flows

Flow	Trigger condition	System behavior	Resulting state
Success — create	Valid input, authorized user	Order saved as `Draft` with new order number	SCR-001 edit mode for the new order, success banner
Success — edit	Valid input and transition	Changes saved	SCR-001 edit mode with saved values, success banner
Exception — unauthenticated	No valid session (including expiry mid-edit)	401, nothing saved	Redirect to `/login`
Exception — forbidden	Signed in without `Admin` / `Operator` role	403, nothing saved	Error panel 「製造指示を管理する権限がありません。」 (You don't have permission to manage production orders.)
Exception — validation	Any V-01–V-05 fails	Rejected, nothing saved	Inline field errors; values kept
Exception — invalid transition	V-06 fails (e.g. crafted request)	Rejected, nothing saved	Error banner; status unchanged
Exception — locked field changed	V-07 fails	Whole request rejected (DEC-007)	Error banner; values kept
Exception — order not found	Edit route/API with unknown `id`	404	Not-found panel with a link back to the list
Exception — products unavailable	Product list empty or fails to load	Save not possible in create mode	Product select shows 「利用可能な製品がありません。」 (No products available.) / load error banner
Exception — concurrent change	V-08 fails: order was changed by someone else after it was loaded	Whole request rejected, nothing saved (DEC-010)	Error banner MSG-E009 with a Reload action; entered values kept until reload
Exception — server/network error	Unexpected failure	Nothing assumed saved	Error banner 「問題が発生しました。もう一度お試しください。」 (Something went wrong. Try again.); values kept

Data design overview

- **ProductionOrder** — order number (system-generated, unique, `PO-YYYY-NNNNN`), reference to exactly one Product, quantity, due date, status (`Draft` / `InProgress` / `Completed` / `Cancelled`), notes, created/updated timestamps, and the completion time — present exactly when the status is `Completed` (added by WI-004, 003_DB).
- **Product** — name, SKU; a minimal reference table seeded with 30 sample automobile parts with Japanese names, such as `P-1001` ブレーキキャリパー (brake caliper), no catalog UI (DEC-005, DEC-019; WI-005 DEC-002).

Relationship: many ProductionOrders → one Product (required). Column types, keys, constraints and order-number generation belong to the database-design document.

External interfaces

None — only this application's own backend API.

Non-functional requirements

- **Security:** the role check (`Admin` / `Operator`) gates state-changing actions and must be enforced server-side. No field is PII or a secret. CSRF: no extra anti-forgery token; `SameSite=Lax` cookie, no CORS policy and JSON-only request bodies (DEC-020).
- **Accessibility (WCAG 2.2 AA, `ai/rules/frontend.md`):** every control has a visible label; required fields use `aria-required` ; inline errors are linked via `aria-describedby` ; the message banner uses a live region; focus moves to the first invalid field on failed save; the discard-changes dialog traps focus, starts on **編集を続ける** (Keep editing) and closes on Escape; read-only (locked) fields remain focusable and announced as read-only; visible focus indicators and sufficient contrast; fully keyboard-operable (including the date picker).
- **Observability:** create/edit endpoints are traced and logged per project defaults; exact spans/metrics are specified in 001_DD.
- **Language:** the UI is Japanese only (`<html lang="ja">` , Japanese font stack); every text comes from the frontend catalog (WI-005 DEC-001, DEC-007).
- Performance/availability: inherits project defaults.

Open questions and linked DD

Question	Linked DD section	Status
Does the due-date ≥ today rule apply when an existing order's past due date is left unchanged?	001_DD validation	answered in decisions.md (DEC-009)
Concurrent edits: reject stale save or last write wins?	001_DD processing flow / API contract	answered in decisions.md (DEC-010); mechanism decided in 001_DB (DEC-014)
Which timezone defines "today"?	001_DD validation / configuration	answered in decisions.md (DEC-011)
Order-number format	Database design (001_DB)	answered in decisions.md (DEC-012); generation and overflow handling decided in 001_DB (DEC-013)
CSRF protection for the new endpoints	001_DD security	answered in decisions.md (DEC-020)

Question	Linked DD section	Status
Plant timezone value; Cancel with unsaved changes; demo products	001_DD configuration / events; 001_DB seed	answered in decisions.md (DEC-017–DEC-019)